

FAKEAGENT / SECTOPRAT DLL SIDELOADING

Signed-application DLL sideloading through source-observed process-module pairs.

HUNTING**CONFIDENCE: HIGH**

FAKEAGENT / SECTOPRAT

Identify the exact source-supported process-module pairs without using volatile domains, IP addresses or hashes as blocking conditions.

MALVERTISING

FAKE INSTALLER

SIGNED BINARY LAUNCH

DLL SIDELOAD

SECTOPRAT ACTIVITY

REPORT DATE**23 July 2026****VERSION****v2.0****PRIMARY TELEMETRY****Microsoft Defender XDR ·
DeviceImageLoadEvents****ATT&CK / SCOPE****T1574.002 · T1036**

Independent defensive research. Source facts, analytical assessment, assumptions and detection logic are intentionally separated. Every query requires environment-specific validation.

THREAT RESEARCH EDITORIAL STANDARD · HTML MASTER V2

01

EXECUTIVE ASSESSMENT

Decision-grade summary for threat research, detection engineering and SOC operations.

ASSESSMENT

EXECUTIVE SUMMARY

FakeAgent / SectopRAT DLL Sideloads

Huntress documented renamed signed executables loading co-located DLLs: ClaudeDesktop.exe or DockerDesktop.exe loading libcef.dll, and sslconf.exe loading tempdir.dll.

WHY IT MATTERS

Operational detection value

The process-module pairs are directly supported by primary research and observable through image-load telemetry. A match still requires provenance, signer, original filename and follow-on analysis.

High 01

ANALYTICAL CONFIDENCE



Focused 02

TELEMETRY COVERAGE



Hunting 03

OPERATIONAL READINESS



SOURCE-CONFIRMED FACT

Documented behavior

The campaign used the specified renamed executable and DLL pairings across multiple observed organizations.

ANALYTICAL ASSESSMENT

Interpretation

The pairings are strong campaign hunting anchors, but they do not independently establish that the loaded DLL is modified or malicious.

DETECTION ASSUMPTION

Required condition

DeviceImageLoadEvents captures both process and module fields consistently across the target fleet.

Evidence boundary

It does not prove DLL tampering, SectopRAT execution, delivery method, persistence or attribution.

02

THREAT MECHANICS

Behavioral decomposition and observable evidence boundaries.

ATTACK FLOW



STAGE-BY-STAGE DETECTION VALUE

STAGE	BEHAVIOR	EVIDENCE EXPECTATION	USE
01	Malvertising	Supporting process, file, identity or application context must be collected and correlated.	Context / enrichment
02	Fake installer	Supporting process, file, identity or application context must be collected and correlated.	Context / enrichment
03	Signed binary launch	The campaign used the specified renamed executable and DLL pairings across multiple observed organizations.	Primary anchor
04	DLL sideload	Supporting process, file, identity or application context must be collected and correlated.	Context / enrichment
05	SectopRAT activity	Supporting process, file, identity or application context must be collected and correlated.	Context / enrichment

Observable core

A match indicates that a source-observed executable name loaded the corresponding campaign DLL name.

Non-observable boundary

It does not prove DLL tampering, SectopRAT execution, delivery method, persistence or attribution.

TELEMETRY & EVIDENCE MODEL

Minimum data contract required for a defensible implementation.

DATA CONTRACT

PRIMARY DATA SOURCE

Microsoft Defender XDR · DeviceImageLoadEvents

The primary platform is selected because it observes the strongest source-supported behavior with the most direct event relationship.

EVIDENCE BOUNDARY

A match is an investigative lead.

It does not prove DLL tampering, SectopRAT execution, delivery method, persistence or attribution.

REQUIRED FIELD CONTRACT

TIER	DATA	PURPOSE	RELIABILITY
REQUIRED	Timestamp, DeviceId, DeviceName, InitiatingProcessFileName, InitiatingProcessFolderPath, InitiatingProcessCommandLine, FileName, FolderPath, SHA1, InitiatingProcessSHA1, ReportId	Core filtering, grouping and investigation output.	Documented schema; population must be verified.
OPTIONAL	Signer, hash, prevalence, file origin, software inventory, device group.	Benign separation and prioritization.	Environment-dependent.
ENRICHMENT	Threat intelligence, change windows, asset criticality and identity context.	Disposition and escalation.	Integration-dependent.

Completeness

Confirm table coverage, retention, field population and event latency before measuring prevalence.

Normalization

Validate case, path, quoting, URI encoding and command-line preservation against raw events.

Visibility gap

Renamed files and alternate loaders evade the narrow logic; generic libcef.dll activity is too common.

04

DETECTION LOGIC

Source-grounded implementation with explicit assumptions and outcome boundaries.

HUNTING

DETECTION OBJECTIVE

Identify the exact source-supported process-module pairs without using volatile domains, IP addresses or hashes as blocking conditions.

IMPLEMENTATION STATUS

Hunting - Untested implementation sketch

Syntax and schema are documented; local execution, field population and false-positive performance are not established.

KQL · DEMONSTRATIVE

```
// Title: FakeAgent Signed-Application DLL Sideloadig
// Status: Hunting query - Untested implementation sketch
// Confidence: High
// Platform: Microsoft Defender XDR
// Telemetry: DeviceImageLoadEvents
// Purpose: Hunt for the process-module pairs observed by Huntress in the
// FakeAgent campaign, renamed JetBrains components loading libcef.dll and an
// IBM SPSS sslconf.exe binary loading tempdir.dll.
// ATT&CK: T1574.002, T1036
//
// Evidence boundary:
// - Each process-module pair is directly supported by the primary source.
// - A match does not prove that the loaded DLL is modified or malicious.
// - File provenance, original filename, signature, prevalence, persistence and
//   follow-on endpoint activity must be reviewed before escalation.
//
// Validation:
// Run over 30-90 days and group distinct process paths, module paths, hashes,
// devices and software owners. Keep as hunting unless benign activity can be
// separated through stable, narrowly scoped attributes.

DeviceImageLoadEvents
| where
(
    InitiatingProcessFileName in~ ("ClaudeDesktop.exe", "DockerDesktop.exe")
    and FileName =~ "libcef.dll"
)
or
(
    InitiatingProcessFileName =~ "sslconf.exe"
    and FileName =~ "tempdir.dll"
)
| project
    Timestamp,
    DeviceId,
    DeviceName,
    InitiatingProcessFileName,
    InitiatingProcessFolderPath,
    InitiatingProcessCommandLine,
    FileName,
    FolderPath,
    SHA1,
    InitiatingProcessSHA1,
    ReportId
```

What a match means

A match indicates that a source-observed executable name loaded the corresponding campaign DLL name.

What a match does not prove

It does not prove DLL tampering, SectopRAT execution, delivery method, persistence or attribution.

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TRIAGE & INVESTIGATION

Evidence-led workflow for defensible disposition.

SOC WORKFLOW

NUMBERED WORKFLOW

- 01

Confirm the match.
Review the raw event, timestamp, entity relationship and exact matching fields.
- 02

Establish legitimacy.
Identify the software or change owner, expected path, signer and authorized maintenance window.
- 03

Reconstruct origin.
Pivot to process, file, identity, network or application events before and after the match.
- 04

Measure prevalence.
Compare hashes, paths, commands, users and devices across the retained fleet.
- 05

Document confidence.
Separate confirmed evidence, inference, assumptions and unresolved gaps.

DECISION MODEL

Approved context → tune narrowly

Known owner, expected software path and repeatable behavior can support a documented exclusion.

Unknown context → deepen review

Rare or risky context without corroboration remains suspicious or insufficient evidence.

Corroborated activity → escalate

Malicious follow-on process, file, network, persistence or identity evidence supports incident escalation.

Host

Coverage, exposure, recent changes and neighboring alerts.

User

Session, intent, role and authorization context.

Process

Parent, child, path, command, signer and original filename.

File

Hash, signature, origin, prevalence and creation timeline.

Network

DNS, proxy, TLS and destinations following the event.

Identity

Sign-ins, token use, privilege and resource access.

OUTCOME	EVIDENCE PROFILE	ACTION
Benign	Approved and repeatable behavior with strong ownership.	Document and tune narrowly.
Suspicious	Risky context without decisive malicious corroboration.	Deepen endpoint or application review.
Likely malicious	Source-compatible match plus corroborated malicious activity.	Escalate incident response.
Insufficient evidence	Missing telemetry or unresolved ownership.	Retain evidence and validate gaps.

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FALSE POSITIVES & VALIDATION

Operational controls required before alerting.

QUALITY GATE

FALSE-POSITIVE MATRIX

SCENARIO	LIKELIHOOD	TUNING ACTION
Legitimate CEF application packaging	High	Review owner, path, signer, prevalence and change context before exclusion.
IBM SPSS components	High	Review owner, path, signer, prevalence and change context before exclusion.
Software repackaging	Med	Review owner, path, signer, prevalence and change context before exclusion.
Security simulation	Med	Review owner, path, signer, prevalence and change context before exclusion.

Baseline plan

- Run retrospectively over a representative retention window.
- Measure volume by device, user, process, path and software owner.
- Inspect recurring legitimate patterns before adding exclusions.
- Preserve raw examples for peer review.

Validation procedure

- Confirm every required field against raw events.
- Execute only safe, authorized lab or historical validation.
- Compare query output with endpoint or application ground truth.
- Record success, rejection and sensor-failure criteria.

PROMOTION GATE

Hunting

Source-grounded hypothesis.

Candidate

Schema and baseline complete.

Production

Measured precision and runbook.

Optimized

Stable feedback loop.

Promotion criteria

Known alert volume, documented exclusions, representative validation, triage ownership, data-quality monitoring and measurable precision.

Failure condition

Renamed files and alternate loaders evade the narrow logic; generic libcef.dll activity is too common.

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ATT&CK, CONCLUSION & REFERENCES

Coverage statement, residual risk and document control.

CLOSURE

COVERAGE MAPPING

MAPPING	COVERAGE STATEMENT	BOUNDARY
T1574.002 · T1036	Partial behavioral or campaign-artifact coverage of the source-supported observable.	It does not prove DLL tampering, SectopRAT execution, delivery method, persistence or attribution.

ANALYST CONCLUSION

Assessment, not certainty

The process-module pairings are assessed as a high-confidence hunting lead pending baseline, provenance and follow-on behavioral confirmation.

Residual risk

Renamed files and alternate loaders evade the narrow logic; generic libcef.dll activity is too common.

REFERENCES

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DOCUMENT CONTROL

VERSION	DATE	STATUS	OWNER
v2.0	23 July 2026	Hunting	Detection Engineering

Change summary

Migrated to the permanent Threat Research Editorial HTML master; technical query and evidence classification preserved.

Publication control

No customer data, credentials or environment-specific internal indicators are included. URLs are defanged in the publication artefact.